

1. Consider the system defined by the following equations:

$$\dot{x}_1 = \frac{2}{3}x_2$$

$$\dot{x}_2 = -x_1 + x_2(1 - 3x_1^2 - 2x_2^2)$$

(a) Show that the points defined by (i) $x = (0, 0)$ and (ii) $1 - (3x_1^2 + 2x_2^2) = 0$ are invariant sets.

(b) Study the stability of the origin and the invariant set $1 - (3x_1^2 + 2x_2^2) = 0$, respectively, using LaSalle's Invariant Theorem.

1. (a) (i) $x = (0, 0)$, $\dot{x}_1 = 0$, $\dot{x}_2 = 0$,

then $x_1 = x_2 = 0$, $\forall t \geq 0$, and then $x = (0, 0)$ is an invariant set.

(ii) $1 - (3x_1^2 + 2x_2^2) = 0 \Rightarrow \dot{x}_1 = \frac{2}{3}x_2$, $\dot{x}_2 = -x_1$

the time derivative of $V = 3x_1^2 + 2x_2^2$ is given by

$$\dot{V} = -6x_1 \cdot \dot{x}_1 - 4x_2 \cdot \dot{x}_2 = -6x_1 \cdot \frac{2}{3}x_2 - 4x_2(-x_1) = 0$$

that means V is constant. $V = V_0 = 1$, $3x_1^2 + 2x_2^2 = 1$, then $3x_1^2 + 2x_2^2 = 1$ is an invariant set.

(b) consider the function $V = \frac{3}{4}x_1^2 + \frac{1}{2}x_2^2$

$$\dot{V} = \frac{3}{2}x_1\dot{x}_1 + x_2\dot{x}_2 = x_1x_2 + x_2[-x_1 + x_2(1 - 3x_1^2 - 2x_2^2)]$$

$$= x_2^2(1 - 3x_1^2 - 2x_2^2) > 0 \text{ for the neighborhood of the origin}$$

$\Rightarrow$ the origin is not stable.

And then investigate the stability of $3x_1^2 + 2x_2^2 = 1$

$$V(x) = \frac{1}{4}(3x_1^2 + 2x_2^2 - 1)^2$$

$$\dot{V}(x) = \frac{1}{2}(3x_1^2 + 2x_2^2 - 1)(6x_1\dot{x}_1 + 4x_2\dot{x}_2)$$

$$= (3x_1^2 + 2x_2^2 - 1)(2x_2x_1 + 2x_2[-x_1 + x_2(1 - 3x_1^2 - 2x_2^2)])$$

$$= -2x_2^2(3x_1^2 + 2x_2^2 - 1)^2 \leq 0$$

Step 1. $\Omega_c = \{x \in \mathbb{R}^2 \mid V(x) \leq c\}$ is a invariant set.

Step 2 $E = \{x \in \Omega_c \mid \dot{V}(x) = 0\} = \{x \mid 3x_1^2 + 2x_2^2 - 1 = 0\} \cup \{x \mid x_2 = 0\}$, $c \geq 1$

Step 3 Find M , the largest invariant set in E

$$M = \{x \mid 3x_1^2 + 2x_2^2 - 1 = 0\} \cup \{(0, 0)\}$$

choose $c \in [0, 1)$, Ω_c include the cycle but not the origin. and the LaSalle's theorem for $c \in [0, 1)$ shows that every motion starting in Ω_c converges to the limit cycle, and then the limit cycle $\{x \mid 3x_1^2 + 2x_2^2 - 1 = 0\}$ is stable.

And choose $c = 1 - \varepsilon$, $\varepsilon \neq 0$, all motion leads to the cycle, and the origin is unstable.

2. It is known that a given dynamical system with the state $x = (x_1, x_2)$ has an equilibrium point at the origin. For this system, a function $V(\cdot)$ have been proposed, and its derivative $\dot{V}(\cdot)$ has been computed. Assuming that $V(\cdot)$ and $\dot{V}(\cdot)$ are given below you are asked to classify the origin, in each case, as (a) stable, (b) locally uniformly asymptotically stable, and/or (c) globally uniformly asymptotically stable. Explain your answer in each case.

- (i) $V(x, t) = x_1^2 + x_2^2, \dot{V}(x, t) = -x_1^2.$
- (ii) $V(x, t) = x_1^2 + x_2^2, \dot{V}(x, t) = -(x_1^2 + x_2^2)e^{-t}.$
- (iii) $V(x, t) = x_1^2 + x_2^2, \dot{V}(x, t) = -(x_1^2 + x_2^2)e^t.$
- (iv) $V(x, t) = (x_1^2 + x_2^2)e^t, \dot{V}(x, t) = -(x_1^2 + x_2^2)(1 + \sin^2 t).$
- (v) $V(x, t) = (x_1^2 + x_2^2)e^{-t}, \dot{V}(x, t) = -(x_1^2 + x_2^2).$
- (vi) $V(x, t) = (x_1^2 + x_2^2)(1 + e^{-t}), \dot{V}(x, t) = -x_1^2 e^{-t}.$
- (vii) $V(x, t) = (x_1^2 + x_2^2)(1 + \cos^2 t), \dot{V}(x, t) = -(x_1^2 + x_2^2)e^{-t}.$
- (viii) $V(x, t) = (x_1^2 + x_2^2)(1 + \cos^2 t), \dot{V}(x, t) = -(x_1^2 + x_2^2)(1 + e^{-t}).$

(i) $W_1 = V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d., $\dot{V} \leq 0$.
then the origin is uniformly stable.

(ii) $W_1 = V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d., $\dot{V} \leq 0$,
then the origin is uniformly stable.

(iii) $W_1 = V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d. $\dot{V}(x, t) \leq -(x_1^2 + x_2^2) = -W_3(x)$
 $W_3(x)$ is p.d. and for $D = \mathbb{R}^2$, $W_1(x)$ is radially unbounded, then $x=0$ is (c) GUAS.

(iv) can't find p.d. function $W_2(x)$ such that $V \leq W_2(x)$
then $x=0$ is not uniformly stable.

(v) can't find a p.d. function $W_1(x)$ such that $W_1(x) \leq V$
can't consider the stability of $x=0$.

(vi) $W_1(x) = (x_1^2 + x_2^2) \leq V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d. $\dot{V} \leq 0$
then $x=0$ is uniformly stable.

(vii) $W_1(x) = (x_1^2 + x_2^2) \leq V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d. $\dot{V} \leq 0$
then $x=0$ is uniformly stable.

(viii) $W_1(x) = (x_1^2 + x_2^2) \leq V \leq W_2 = 2(x_1^2 + x_2^2)$, W_1, W_2 are p.d. $\dot{V} \leq -(x_1^2 + x_2^2) = -W_3(x)$
 $W_3(x)$ is p.d. and $W_1(x)$ is radially unbounded, and then $x=0$ is (c) GUAS.

3. A pendulum with time-varying friction is represented by

$$\dot{x}_1 = x_2, \quad (1)$$

$$\dot{x}_2 = -\sin x_1 - g(t)x_2. \quad (2)$$

Suppose that $g(t)$ is continuously differentiable and satisfies

$$0 < a < \alpha \leq g(t) \leq \beta < \infty \quad \text{and} \quad \dot{g}(t) \leq \gamma < 2$$

for all $t \geq 0$. Consider the Lyapunov function candidate

$$V(t, x) = \frac{1}{2}(a \sin x_1 + x_2)^2 + [1 + ag(t) - a^2](1 - \cos x_1) \quad (3)$$

(a) Show that $V(t, x)$ is positive definite and decrescent.

(b) Show that

$$\dot{V} \leq -(\alpha - a)x_2^2 - a(2 - \gamma)(1 - \cos x_1) + O(\|x\|^3), \quad (4)$$

where $O(\|x\|^3)$ is a term bounded by $k\|x\|^3$ in some neighborhood of the origin.

(c) Show that the origin is uniformly asymptotically stable.

$$\begin{aligned} (a) \quad V(t, x) &= \frac{1}{2}(a \sin x_1 + x_2)^2 + 2 \sin^2 \frac{x_1}{2} (1 + ag(t) - a^2) \\ &> \frac{1}{2}(a \sin x_1 + x_2)^2 + 2 \sin^2 \frac{x_1}{2} (1 + a \cdot a - a^2) \\ &= \frac{1}{2}(a \sin x_1 + x_2)^2 + 2 \sin^2 \frac{x_1}{2} = W_1(x) \end{aligned}$$

Let $W_1(x) = 0 \Rightarrow a \sin x_1 + x_2 = 0, \sin^2 \frac{x_1}{2} = 0, x_1 \in (-\pi, \pi) \Rightarrow x_1 = x_2 = 0$

so $W_1(x)$ is p.d. and then $V(t, x)$ is positive definite.

$$V(t, x) \leq \frac{1}{2}(a \sin x_1 + x_2)^2 + 2 \sin^2 \frac{x_1}{2} (1 + a\beta - a^2) = W_2(x)$$

$1 + a\beta - a^2 > 0$ and $W_2(x)$ is p.d. and then $V(t, x)$ is decrescent.

$$\begin{aligned} (b) \quad \dot{V} &= \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x_1} \cdot \dot{x}_1 + \frac{\partial V}{\partial x_2} \cdot \dot{x}_2 \\ &= a \dot{g}(t)(1 - \cos x_1) + (a \sin x_1 + x_2) \cdot a \cos x_1 + [1 + ag(t) - a^2] \sin x_1 \cdot x_2 \\ &\quad + (a \sin x_1 + x_2)(-\sin x_1 - g(t)x_2) \\ &= \underbrace{a \dot{g}(t)(1 - \cos x_1)} + \underbrace{x_2 a^2 \sin x_1 \cos x_1} + \underbrace{x_2^2 a \cos x_1} + \underbrace{x_2 \sin x_1}_{\cancel{x_2 \sin x_1}} \cdot \underbrace{ag(t)}_{\cancel{ag(t)}} \\ &\quad - \underbrace{a^2 x_2 \sin x_1}_{\cancel{a^2 x_2 \sin x_1}} - \underbrace{a \sin^2 x_1}_{\cancel{a \sin^2 x_1}} - \underbrace{ag(t) x_2 \sin x_1}_{\cancel{ag(t) x_2 \sin x_1}} - \underbrace{x_2^2 \sin x_1}_{\cancel{x_2^2 \sin x_1}} - \underbrace{x_2^2 g(t)}_{\cancel{x_2^2 g(t)}} \\ &= -a[1 + \cos x_1 - \dot{g}(t)](1 - \cos x_1) + [a \cos x_1 - g(t)]x_2^2 + a^2 x_2 \sin x_1 (\cos x_1 - 1) \\ &= -a[2 + \cos x_1 - 1 - \dot{g}(t)](1 - \cos x_1) + [a \cos x_1 - g(t)]x_2^2 + a^2 x_2 \sin x_1 (\cos x_1 - 1) \\ &= -a[2 - \dot{g}(t)](1 - \cos x_1) + [a \cos x_1 - g(t)]x_2^2 + a(a x_2 \sin x_1 + \cos x_1 - 1)(\cos x_1 - 1) \\ &\leq -a(2 - \gamma)(1 - \cos x_1) - (\alpha - a)x_2^2 + R_n(x) \\ &\because \sin x \sim x, \cos x - 1 \sim \frac{1}{2}x^2 \therefore R_n(x) = a(a x_2 \sin x_1 + \cos x_1 - 1)(\cos x_1 - 1) \sim O(x^4) \\ &\therefore R_n(x) \leq O(\|x\|^3) \\ &\therefore \dot{V} \leq -(\alpha - a)x_2^2 - a(2 - \gamma)(1 - \cos x_1) + O(\|x\|^3) \end{aligned}$$

$$cc) \because \dot{V} \leq W(x) + R_n(x) = -a(2-t)(1-\cos x_1) - (d-a)x_2^2 + R_n(x)$$

$W(x)$ is N.D and $R_n(x) \sim O(x^4)$, $\frac{O(x^4)}{O(x^2)} \rightarrow 0$
and then V is p.d and decrescent.

Ultimately the system is uniformly asymptotically stable.

4. (1) Barbalat's lemma:

If the differentiable function $f(t)$ has a finite limit as $t \rightarrow \infty$,
and if $\dot{f}(t)$ is uniformly continuous, then $\lim_{t \rightarrow \infty} \dot{f}(t) = 0$

"Lyapunov-Like" Lemma:

If a scalar function $V(t, x)$ satisfies the following conditions,

(1) $V(t, x)$ is lower bounded

(2) $\dot{V}(t, x)$ is negative semidefinite

(3) $\dot{V}(t, x)$ is uniformly continuous

and then $\lim_{t \rightarrow \infty} \dot{V}(t, x) = 0$

Lashella-Yoshizawa theorem:

Let $x=0$ be an equilibrium point of $\dot{x} = f(t, x)$ and suppose that $f(t, x)$ is locally Lipschitz in x and uniformly in t . Let $V(t, x)$ be a continuously differentiable function such that

$$d_1(|x|) \leq V(t, x) \leq d_2(|x|)$$

$$\dot{V} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \leq -W(x) \leq 0$$

$\forall t \geq 0, x \in \mathbb{R}^n$, where $d_1(\cdot)$ and $d_2(\cdot)$ are class k_∞ functions and $W(x)$ is a continuous function. Then all solutions of $\dot{x} = f(t, x)$ satisfy

$$\lim_{t \rightarrow \infty} W(x(t)) = 0$$

In addition, if $W(x)$ is p.d., $x=0$ is globally uniformly asymptotically stable.

(2)(i) use Barbalat's lemma to proof "Lyapunov-like" lemma:

$V(t,x)$ is lower bound and $\dot{V}(t,x)$ is negative semidefinite and then $V(t,x)$ has a finite limit

Moreover, since $\dot{V}(t,x)$ is uniformly continuous, and through Barbalat's lemma, then $\lim_{t \rightarrow \infty} \dot{V}(t,x) = 0$

(ii) use Barbalat's lemma to proof "LaSalle-Yoshizawa" theorem:

if $\int_0^t w(x(\tau)) d\tau$ has a finite limit, $w(x(t))$ is uniformly $\Rightarrow w(x(t)) \rightarrow 0$

And note $w(x) \geq 0$ $\int_0^t w(x(\tau)) d\tau$ is increasing

From $\dot{V} \leq -w(x)$, $V(t) - V(0) \leq -\int_0^t w(x(\tau)) d\tau$

$\Rightarrow \int_0^t w(x(\tau)) d\tau \leq V(0) - V(t) \leq V(0)$, thus $\int_0^t w(x(\tau)) d\tau$ has a finite limit.

Since $\dot{V} \leq 0$, $\alpha_1(\|x\|) \leq V(t,x) \leq \alpha_2(\|x\|) \Rightarrow V$ is bounded

$\|x\| \leq \alpha_1^{-1}(V(t,x))$ is bounded

Assume that $\|x\| \leq R (\triangleq \alpha_1^{-1}(V(0)))$, Note that a continuous function is uniformly continuous on a closed set, $w(x)$ is uniformly continuous in x
 $\forall t_2 > t_1 \geq 0 \quad |x(t_2) - x(t_1)| \leq k \cdot (t_2 - t_1)$

$$|x(t_2) - x(t_1)| = \left| \int_{t_1}^{t_2} f(\tau, x(\tau)) d\tau \right| \leq LR \int_{t_1}^{t_2} \|x(\tau)\| d\tau \leq LR R (t_2 - t_1) \quad (\|x\| \leq R)$$

Then $x(t)$ is uniformly continuous in t , and thus $w(x(t))$ is also uniformly continuous in t . From Barbalat's lemma, $\lim_{t \rightarrow \infty} w(x(t)) = 0$

(iii) use Barbalat's lemma to proof Corollary 0.1

Let $g(t) = \int_0^t x^2(\tau) d\tau$, $g(t)$ has a finite limit.

$$\dot{g}(t) = x^2, \quad \ddot{g}(t) = 2x \cdot \dot{x} \in L_\infty$$

From Barbalat's lemma, $\lim_{t \rightarrow \infty} \dot{g}(t) = \lim_{t \rightarrow \infty} x^2(t) = 0$

hence, $\lim_{t \rightarrow \infty} x(t) = 0$

5. Consider the following multi-dimensional system

$$\dot{x} = Ax + B(u + \Theta^T \Phi(x))$$

where $x \in \mathbb{R}^n$ is the state, $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{m \times n}$ are known matrices, $u \in \mathbb{R}^m$ is the control input, $\Phi(x) \in \mathbb{R}^k$ is a bounded function, and $\Theta \in \mathbb{R}^{k \times m}$ is an unknown constant matrix. Assume that (A, B) is controllable.

- Design an adaptive control law to stabilize the system.
- Design an adaptive control law with adaptive σ -modification to stabilize the system.

(1) We can find A_{ref} , let $\dot{x}_{ref} = A_{ref} x_{ref}$, A_{ref} is Hurwitz.

and design $u = k_1^* x + k_2^* u_c - \theta^T \phi(x)$

$$\begin{cases} A_{ref} = A + B k_1^* \\ B k_2^* = 0 \end{cases} \quad A_{ref}, A, B \Rightarrow k_1^*, k_2^*$$

$$u = k_1^* x + k_2^* u_c - \hat{\theta}(t)^T \phi(x)$$

$$\Rightarrow \dot{x} = A_{ref} x - B \tilde{\theta}^T \phi(x)$$

$$\text{Consider } V = x^T P x + \text{tr}(\tilde{\theta} \tilde{\theta}^T) \geq 0$$

where P is confirmed by $A_{ref}^T P + P A_{ref} = -Q < 0$, Q is p.d.

$$\dot{V} = (x^T A_{ref}^T - \phi^T \tilde{\theta} B^T) P x + x^T P (A_{ref} x - B \tilde{\theta}^T \phi(x)) + 2 \text{tr}(\dot{\tilde{\theta}} \tilde{\theta}^T)$$

$$= x^T (A_{ref}^T P + P A_{ref}) x - (\phi^T \tilde{\theta} B^T P x + x^T P B \tilde{\theta}^T \phi) + 2 \text{tr}(\dot{\tilde{\theta}} \tilde{\theta}^T)$$

$$= -x^T Q x - 2 \phi^T \tilde{\theta} B^T P x + 2 \text{tr}(\dot{\tilde{\theta}} \tilde{\theta}^T)$$

$$= -x^T Q x - 2 \text{tr}(\phi x^T P B \tilde{\theta}^T) + 2 \text{tr}(\dot{\tilde{\theta}} \tilde{\theta}^T)$$

$$\text{Let } \dot{\tilde{\theta}}(t) = \phi x^T P B$$

$$\Rightarrow \dot{V} = -x^T Q x \leq 0 \quad V(x) \leq V(0) \quad \therefore \tilde{\theta}(t), x \in L_\infty \quad (\exists c, c \geq \text{tr}(\tilde{\theta} \tilde{\theta}^T))$$

$\therefore x^T P x \leq V \leq x^T P x + c$, and from Lasalle-Yashizawa theorem, $\lim_{t \rightarrow \infty} x(t) = 0$

$$(2) \quad \begin{cases} \dot{\hat{\theta}} = \phi x^T P B - \sigma(\hat{\theta} - \tilde{\theta}) \\ \dot{\hat{\theta}}_1 = \sigma(\hat{\theta} - \tilde{\theta}) \end{cases} \quad \sigma > 0$$

$$V = x^T P x + \text{tr}(\tilde{\theta} \tilde{\theta}^T) + \text{tr}(\tilde{\theta}_1^T \tilde{\theta}_1)$$

$$\dot{V} = -x^T Q x - 2 x^T P B \tilde{\theta}^T \phi + 2 \text{tr}(\tilde{\theta}_1^T \dot{\tilde{\theta}}_1) + 2 \text{tr}(\tilde{\theta}^T \dot{\tilde{\theta}})$$

$$= -x^T Q x - 2 x^T P B \tilde{\theta}^T \phi + 2 \text{tr}(\tilde{\theta}_1^T \sigma(\hat{\theta} - \tilde{\theta})) + 2 \text{tr}(\tilde{\theta}^T \phi x^T P B - \tilde{\theta}^T \sigma(\hat{\theta} - \tilde{\theta}))$$

$$= -x^T Q x + 2 \sigma \text{tr}(\tilde{\theta}_1^T (\hat{\theta} - \tilde{\theta})) - 2 \sigma \text{tr}(\tilde{\theta}^T (\hat{\theta} - \tilde{\theta}))$$

$$= -x^T Q x - 2 \sigma \text{tr}((\hat{\theta} - \tilde{\theta}_1)^T (\hat{\theta} - \tilde{\theta}_1))$$

$(\hat{\theta} - \tilde{\theta}_1)^T (\hat{\theta} - \tilde{\theta}_1)$ is p.s.d. and its eigenvalues $\lambda_1 \cdots \lambda_m \geq 0$ and $\text{tr}((\hat{\theta} - \tilde{\theta}_1)^T (\hat{\theta} - \tilde{\theta}_1)) \geq 0$

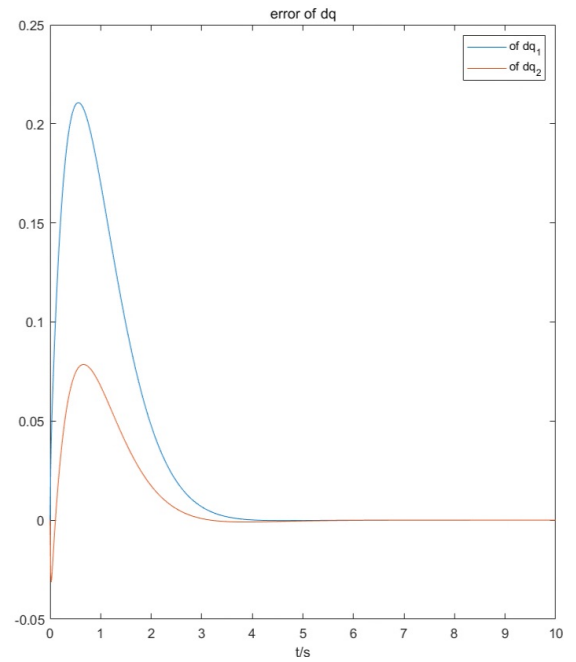
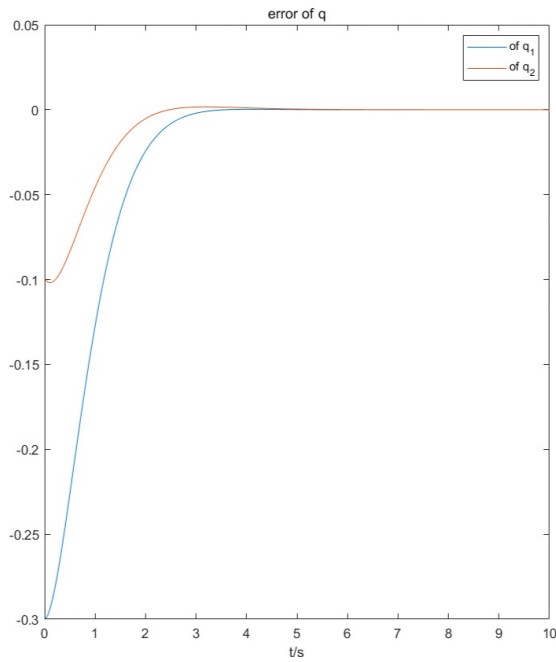
$$\dot{U} \leq -x^T Q x, \quad U(x) \leq U(0)$$

$$\therefore \tilde{\Theta}(t), \tilde{\Theta}_1(t), x \in L_\infty \text{ and then } \exists C > 0, \operatorname{tr}(\tilde{\Theta}^T \tilde{\Theta}) + \operatorname{tr}(\tilde{\Theta}_1^T \tilde{\Theta}_1) \leq C$$

$$\text{and then } x^T P x \leq U \leq x^T P x + C, \quad \alpha_1(\cdot) \leq U \leq \alpha_2(\cdot), \quad \alpha_1(\cdot), \alpha_2(\cdot) \in \mathcal{K}_\infty$$

and from Lasalle Yashizawa theorem, $\lim_{t \rightarrow \infty} x(t) = 0$

6. (a)



(b)

